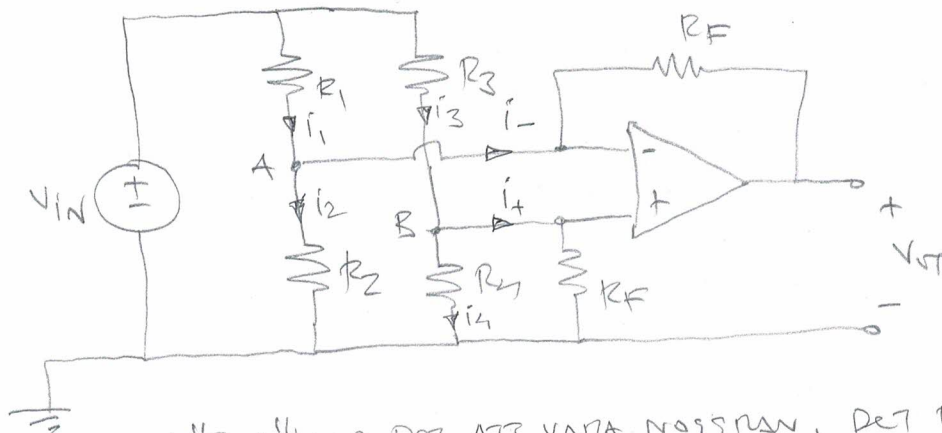


8.38)



HÄR GÄLLER DET ATT VARA NOGSNÄR, DET BLIR MÅNGA
VARIABLER

VI KÖR PÅ MED KSP

$$V_A = R_2 i_2 \quad (1)$$

$$V_B = R_4 i_4 = R_F i_- \quad (2)$$

MEN $V_A = V_B$ FÖR EN IDEAL OP-AMP

$$\Rightarrow R_2 i_2 = R_4 i_4 \quad (3)$$

MEER KSP

$$-V_{OUT} - R_F i_- + R_F i_+ = 0 \quad (4)$$

$$-V_{IN} + R_1 i_1 + R_2 i_2 = 0 \quad (5)$$

$$-V_{IN} + R_3 i_3 + R_4 i_4 = 0 \quad (6)$$

SAMT ATT

$$i_1 - i_2 = i_- \quad (7)$$

$$(3) \Rightarrow i_2 = \frac{R_4}{R_2} i_4 \quad (8)$$

$$(5) \text{ \& } (6) \text{ GER } R_1 i_1 + R_2 i_2 = R_3 i_3 + R_4 i_4$$

$$\text{MED (8)} \quad R_1 i_1 + R_4 i_4 = R_3 i_3 + R_4 i_4$$

$$i_1 = \frac{R_3}{R_1} i_3 \quad (9)$$

MEN

$$i_- = i_1 - i_2 = \frac{R_3}{R_1} i_3 - \frac{R_4}{R_2} i_4$$

odn ATT

$$i_+ = i_3 - i_4$$

(1) qwr R_F

$$-V_{UT} = R_F (i_- - i_+) = R_F \left(\frac{R_3}{R_1} i_3 - \frac{R_4}{R_2} i_4 - i_3 + i_4 \right) =$$

$$= R_F \left[\left(\frac{R_3}{R_1} - 1 \right) i_3 - \left(\frac{R_4}{R_2} - 1 \right) i_4 \right] \quad (10)$$

(2) qwr $i_4 = \frac{R_F}{R_4} i_+ = \frac{R_F}{R_4} (i_3 - i_4)$

$$i_3 = \frac{R_4}{R_F} i_4 + i_4 = \left(1 + \frac{R_4}{R_F} \right) i_4 \quad (11)$$

(11) x (10)

$$-V_{UT} = R_F \left[\left(1 + \frac{R_4}{R_F} \right) \left(\frac{R_3}{R_1} - 1 \right) - \left(\frac{R_4}{R_2} - 1 \right) \right] i_4$$

(6) & (11)

$$-V_{UT} = R_F \left[\left(1 + \frac{R_4}{R_F} \right) \left(\frac{R_3}{R_1} - 1 \right) - \left(\frac{R_4}{R_2} - 1 \right) \right] \frac{V_{IN}}{R_4 + R_3 \left(1 + \frac{R_4}{R_F} \right)}$$

MED SIFFPOR, $R_4 = 10 \text{ K}\Omega$, $R_3 = 15 \text{ K}\Omega$, $R_2 = 10 \text{ K}\Omega$
 $R_1 = 10 \text{ K}\Omega$, $V_{IN} = 6 \text{ V}$, $R_F = 50 \text{ K}\Omega$

$$\Rightarrow -V_{UT} = 50 \left[\left(1 + \frac{10}{50} \right) \left(\frac{15}{10} - 1 \right) - \left(\frac{10}{10} - 1 \right) \right] \frac{6}{10 + 15 \left(1 + \frac{10}{50} \right)}$$

$$\underline{\underline{= +6.428 \text{ V}}}$$

SWAR: $V_{UT} = -6.428 \text{ VOLT}$